## Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

**1** The greatest amount of particles is found in

**A** 5.0 $dm^3$ of $H_2S$ (g) at r.t.p.

**B** $1.8 \times 10^{23}$ molecules of $C_6H_{12}O_6$ (s)

**C** 350 $cm^3$ of 2.5 mol $dm^{-3}$ $HNO_3$ (aq)

**D** 50 g of $Fe_2O_3$(s)

**2** Carbon monoxide, CO, is a colourless, odourless and toxic gas. It is formed as a result of partial oxidation of carbon-containing compounds. The maximum safe toleration level of CO in air is 50ppm. (1ppm = 1mg / kg)

How many molecules of CO gas are present in 1 kg of air at this toleration level?

**A** $\dfrac{50\times10^{-3}\times6.02\times10^{23}}{28}$

**B** $50 \times 10^{-3} \times 28 \times 6.02 \times 10^{23}$

**C** $\dfrac{50\times6.02\times10^{23}}{28}$

**D** $50 \times 28 \times 6.02 \times 10^{23}$

**3** Skunks are mammals best known for their ability to excrete a strong, foul-smelling odour in order to ward off potential attackers. Methanethiol, $CH_3SH$, is a foul-smelling chemical which is found in the sprays of skunks. A 20 $cm^3$ sample of methanethiol was exploded with 80 $cm^3$ of oxygen. What would be the final volume of the resultant mixture of gases when cooled to room temperature and pressure?

$$CH_3SH + 3O_2 \longrightarrow CO_2 + SO_2 + 2H_2O$$

**A** 40

**B** 60

**C** 70

**D** 80

**4.** Which of the following particles would, on losing two electrons, have a half-filled p subshell?

**A** $Ga^-$

**B** $Se^-$

**C** $Te^+$

**D** $As^{2+}$

**5.** The successive ionization energies, in kJ $mol^{-1}$, of an unknown element in period 2 are given below.

900 1760 14800 21000

Which of the following statements about the element is **not** true?

**A** The element is in group II of the Periodic Table.

**B** Its oxide is an amphoteric oxide.

**C** Its chloride has giant ionic structure.

**D** It can conduct electricity in the solid and molten state.

**6.** The bond lengths in buta–1,3–diene differ from those which might be expected.

The carbon – carbon bond length in ethane is 0.154 nm and in ethene 0.134 nm. The central single bond in buta–1,3–diene (C2 – C3), however , is shorter than the single bond in ethane: it is 0.147 nm.

$$\overset{1}{CH_2} \underset{0.134\ nm}{=\!=\!=} \overset{2}{CH} \underset{0.147\ nm}{-\!-\!-} \overset{3}{CH} \underset{0.134\ nm}{=\!=\!=} \overset{4}{CH_2}$$

What helps to explain this C2–C3 bond length?

**A** It is an $sp^2 – sp^2$ overlap.

**B** It is an $sp^2 – sp^3$ overlap.

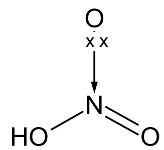
**C** The electrons in the filled p orbitals on C2 and C3 repel each other.

**D** The $sp^3 – sp^3$ bonding is pulled shorter by p–p ( π – bond ) overlap.

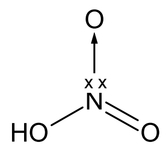
**7** The following diagrams show the electron distribution of $HNO_3$.

Which of the following is an acceptable Lewis diagram of $HNO_3$?

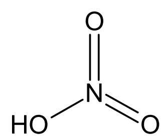
**A**



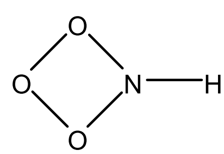
**B**



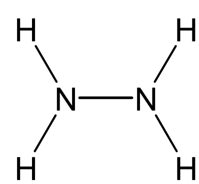
**C**



**D**



**8** Use of the Data Booklet is relevant to this question.
Hydrazine was used as a fuel for the Messerschmidt 163 rocket fighter in World War II and for the American Gemini and Apollo space-craft. It has the following formula.



What is the enthalpy change of atomization of 1 mol of gaseous hydrazine?

**A** 550 kJ $mol^{-1}$

**B** 1720 kJ $mol^{-1}$

**C** 1970 kJ $mol^{-1}$

**D** 2554 kJ $mol^{-1}$

9 A typical protein forms several hundreds of hydrogen bonds and several thousands of Van der Waals' forces in folding from primary to tertiary structures. Which of the following thermodynamic state functions of the protein best represents the folding process?

| | **ΔG / kJ mol$^{-1}$** | **ΔH / kJ mol$^{-1}$** | **ΔS / J K$^{-1}$ mol$^{-1}$** |
|---|---|---|---|
| **A** | < 0 | < 0 | < 0 |
| **B** | < 0 | < 0 | > 0 |
| **C** | > 0 | > 0 | > 0 |
| **D** | > 0 | > 0 | < 0 |

10 The relevant half-equation for the nickel–cadmium cell containing KOH(aq) is as given:

$Cd(OH)_2$ (s) + 2e $\rightleftharpoons$ Cd(s) +2$OH^-$ (aq) $E^\theta = -x$ V

NiO(OH) (s) + $H_2O$ (l) + e $\rightleftharpoons$ $Ni(OH)_2$ (s) + $OH^-$ (aq) $E^\theta = +y$ V

In normal use, electrons are released to the external circuit from the Cd electrode. State which of the following statement is true?

**A** The mass of Cd electrode decreases as current is drawn

**B** Cd is the positive electrode

**C** The concentration of KOH(aq) decreases

**D** The emf of the cell is ($y - x$) V

11 A 0.01 mol sample of a vanadium compound required 20.00 $cm^3$ of 0.1 mol $dm^{-3}$ acidified potassium manganate(VII) for oxidation of the vanadium.

Which one of the following could be the vanadium compound?

**A** $VOF_2$

**B** VOCl

**C** $VO_2F$

**D** $NH_4VO_3$

**12** In acidic solution, 25.0 $cm^3$ of $FeSO_4$ required 20.00 $cm^3$ of 0.100 mol $dm^{-3}$ potassium manganate(VII). In neutral solution, complete oxidation of 25.0$cm^3$ of the same $FeSO_4$ required 33.30 $cm^3$ of 0.100 mol $dm^{-3}$ potassium manganate(VII).

What is the oxidation number of manganese at the end-point?

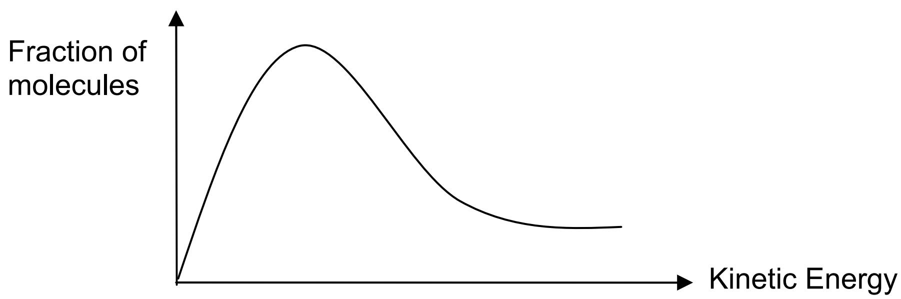
**A** +2

**B** +3

**C** +4

**D** +5

**13** The diagram represents the distribution of molecular kinetic energies within a gas at temperature $T_1$.




Which of the following statements is correct?

**A** The total number of molecules varies at different temperature.

**B** As the gas is cooled, the maximum of the curve will be displaced to the right.

**C** Raising the temperature increases the proportion of molecules **with** any given energy.

**D** At a higher temperature, the proportion of molecules with energies **above** any given value increases.

**14** A gas Q dissociates on heating in a closed vessel to set up the equilibrium as shown below.

$$Q\,(g) \rightleftharpoons R\,(g) + S\,(g)$$

A sample of Q was heated at constant pressure $p$ at a certain temperature. The equilibrium partial pressure of Q was found to be $\frac{1}{7}p$.

What is the equilibrium constant of $K_p$ at this temperature?

**A** $\frac{9}{7}p$

**B** $\frac{6}{7}p$

**C** $\frac{36}{7}p$

**D** $6\,p$

**15** A sample of 1 mol of $N_2O_4$ was placed in an empty vessel and allowed to reach equilibrium according to the equation below.

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

At equilibrium, y% of $N_2O_4$ had dissociated. What is the value of the equilibrium constant, $K_c$, at the temperature of the experiment?

**A** $\frac{2y}{1-y}$

**B** $\frac{4y}{1-y}$

**C** $\frac{4y^2}{100-y} \times \frac{1}{100}$

**D** $\frac{2y}{(100-y)^2} \times \frac{1}{100}$

**16** Some data on two acid-base indicators are shown below

| Indicator | Approximate pH range | Colour Change<br>Acid Alkali |
|---|---|---|
| Bromocresol green | 3.8-5.5 | yellow blue |
| Phenol red | 6.8-8.5 | yellow red |

Which one of the following conclusions can be drawn about a solution in which bromocresol green is blue and phenol red is yellow?

**A** It is strongly acidic

**B** It is weakly acidic

**C** It is neutral

**D** It is weakly alkaline

**17** Which of the following statements about a solution equimolar with respect to both benzoic acid and sodium benzoate is **incorrect**?

**A** It has a pH value equal to the $pK_a$ value of benzoic acid.

**B** Its pH value remains constant when diluted with water.

**C** It can be prepared by mixing 50 $cm^3$ of 0.1 mol $dm^{-3}$ hydrochloric acid with 50 $cm^3$ of 0.2 mol $dm^{-3}$ sodium benzoate.

**D** It can be prepared by mixing 50 $cm^3$ of 0.1 mol $dm^{-3}$ benzoic acid with with 50 $cm^3$ of 0.2 mol $dm^{-3}$ sodium hydroxide.

**18** A solution contains 1 x $10^{-4}$ mol $dm^{-3}$ of magnesium, strontium, iron (II) and silver ions. Which of the following carbonates will be precipitated first when 0.001 mol $dm^{-3}$ of sodium carbonate is added dropwise into the solution?

| | **Compound** | **Solubility product (at 25$^o$C)** |
|---|---|---|
| **A** | Magnesium carbonate | $1.3 \times 10^{-7}$ $mol^2$ $dm^{-6}$ |
| **B** | Strontium carbonate | $9.3 \times 10^{-10}$ $mol^2$ $dm^{-6}$ |
| **C** | Iron(II) carbonate | $2.1 \times 10^{-11}$ $mol^2$ $dm^{-6}$ |
| **D** | Silver carbonate | $8.1 \times 10^{-12}$ $mol^3$ $dm^{-9}$ |

**19** Which of the following is a **true** statement?

**A** $Mg(NO_3)_2$ decomposes to produce $Mg(NO_2)_2$ when heated.

**B** An aqueous solution of $MgCl_2$ is slightly acidic.

**C** Mg reacts more vigorously with water than Ca.

**D** MgO is a yellow solid.

**20** With which substance would solid potassium iodide react to produce iodine?

**A** $Cl_2$ (aq) **B** $H_3PO_4$ (l) **C** HF (g) **D** $AgNO_3$ (aq)

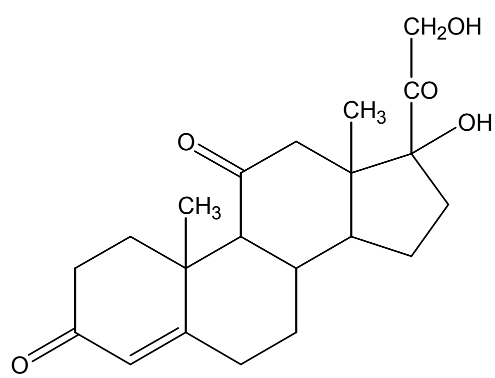
**21** The salt $K_3Fe(CN)_6$ is prepared by oxidizing $K_4Fe(CN)_6$. Which reagent carries out this oxidation?

**A** Ag (s) **B** $Cl_2$ (g) **C** $Cu^{2+}$ (aq) **D** $Fe^{2+}$ (aq)

**22** State the formula of the oxide of titanium that can be used as a white pigment in paint.

**A** TiO **B** $Ti_2O_3$ **C** $TiO_2$ **D** $TiO_3$

**23** The drug cortisone has the formula shown.




How many chiral centres are present in the molecule after its reaction with $LiAlH_4$?

**A** 5 **B** 7 **C** 9 **D** 11

**24** Ethene reacts with aqueous bromine to give $CH_2BrCH_2Br$ and $CH_2BrCH_2OH$. Which statement is correct for the above reaction?

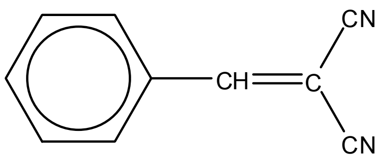
**A** Both products can be hydrolysed to form the same diol.

**B** Both products are non-polar.

**C** Both products are obtained via nucleophilic addition.

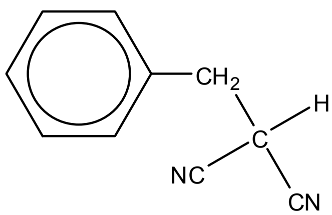
**D** The dibromo product exhibits optical isomerism.

**25** CS, whose structure is shown below, is an active component of 'tear gas' and is readily hydrolysed.

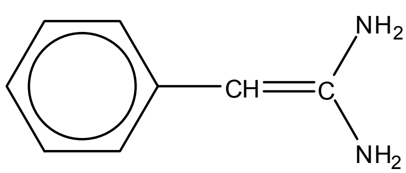


Which of the following is a possible hydrolysis product of CS?

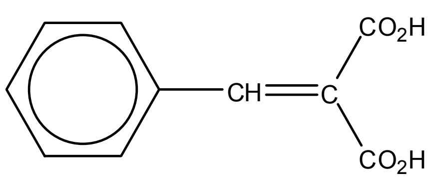
**A**



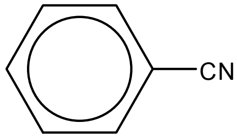
**B**



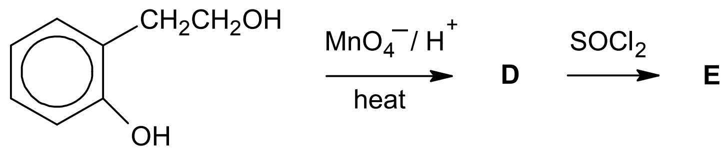
**C**



**D**

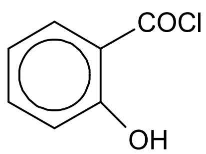


**26** The following compound undergoes the given reaction sequence.

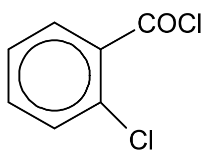


Which of the following could be product **E**?

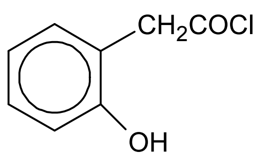
**A**



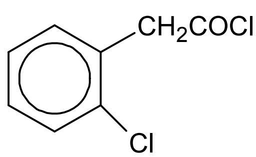
**B**



**C**



**D**



**27** In which sequence is it correctly stated that the value of $pK_a$ **decreases** continuously?

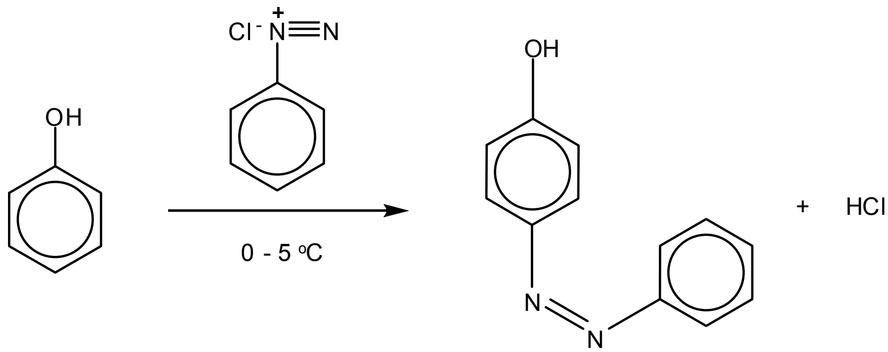
**A** $CH_3CO_2H > CCl_3CO_2H > C_2H_5OH > C_6H_5OH$

**B** $CCl_3CO_2H > CH_3CO_2H > C_2H_5OH > C_6H_5OH$

**C** $C_2H_5OH > C_6H_5OH > CH_3CO_2H > CCl_3CO_2H$

**D** $C_6H_5OH > C_2H_5OH > CH_3CO_2H > CCl_3CO_2H$

**28** Phenol reacts with diazonium chloride as shown.




Which term describes this type of reaction?

**A** electrophilic addition

**B** electrophilic substitution

**C** free-radical substitution

**D** nucleophilic addition

**29** In the study of the polypeptide structure of **A**, it was digested using two different enzymes. The fragments obtained were separated using electrophoresis. Analysis of the fragments from each digestion gave the following results:

| Digestion using the first enzyme: | Digestion using the second enzyme: |
|---|---|
| gly-pro-val | val-asp-gly-thr |
| ser-pro-gly | pro-gly |
| asp-gly | phe-leu-ser |
| thr-phe-leu | gly-pro |

What is the correct sequence of polypeptide structure of **A**?

**A** pro-gly-pro-val-asp-gly- thr-phe-leu-ser-pro-gly

**B** gly-pro-val- ser-pro-gly- asp-gly- thr-phe-leu

**C** gly-pro-val-asp-gly- thr-phe-leu-ser-pro-gly

**D** ser-pro-gly-pro-val-asp-gly- thr-phe-leu-ser

**30** Which of the following reagents may be used to distinguish between methanoic acid and benzaldehyde?

**A** Tollen's reagent

**B** 2,4-dinitrophenylhydrazine

**C** Lithium aluminium hydride

**D** Hydrogen chloride

## Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

| A | B | C | D |
|---|---|---|---|
| **1**, **2** and **3** are correct | **1** and **2** only are correct | **2** and **3** only are correct | **1** only is correct |

No other combination of statements is used as a correct response.

**31** Given that 2 g of a sample of helium ($^4He$) at temperature *T* and volume *V* exerts a pressure *P*. Which of the following would also exert a pressure *P* at the same temperature *T*?

**1** 1 g of hydrogen gas of volume V

**2** A mixture of 0.5 g of hydrogen and 1 g of helium of total volume *V*

**3** A mixture of 1 g of hydrogen and 2 g of helium of total volume 2*V*

**32** Which of the following series show an **increasing** trend in boiling points?

**1** $CH_3CH_2OCH_3$ < $CH_3CH_2CHO$ < $HOCH_2CH_2OH$

**2** $AlF_3$ < $AlBr_3$ < $AlI_3$

**3** $CH_3CH_2CH_2Cl$ < $CH_3CH_2CH_3$ < $CH_3COOH$

**33** Which of the following is always exothermic?

**1** Second electron affinity

**2** Enthalpy change of neutralisation

**3** Lattice energy

**34** The equation of a reaction in equilibrium is shown below:

$$\text{reactants} \underset{k_b}{\overset{k_f}{\rightleftharpoons}} \text{products}$$

$k_f$ = forward rate constant, $k_b$ = backward rate constant

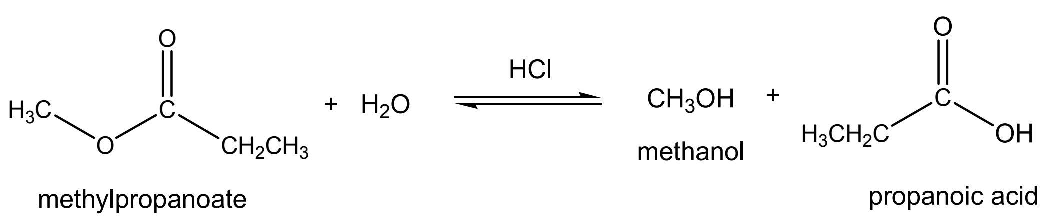
Which of the following changes would cause both the forward and backward rate constants to be increased?

**1** Introduction of a catalyst

**2** Heating the equilibrium mixture

**3** Increasing the concentration of the reactants

**35** For the hydrolysis of methyl propanoate in dilute HCl, the following rates of reactions were measured at 25°C.




| Experiment | [methyl propanoate] / mol dm$^{-3}$ | [HCl] / mol dm$^{-3}$ | Initial rate / mol dm$^{-3}$ min$^{-1}$ |
|---|---|---|---|
| 1 | 0.10 | 0.20 | 0.096 |
| 2 | 0.15 | 0.10 | 0.072 |
| 3 | 0.10 | 0.10 | 0.048 |

Which of the following statements is/are correct?

**1** The rate equation of the reaction is rate = k[methyl propanoate][HCl].

**2** Doubling the concentration of both methyl propanoate and HCl increases the rate of reaction by a factor of 4.

**3** If the half life of methyl propanoate in experiment 1 is 5.0 minutes, then the half life of methyl propanoate in experiment 3 is also 5.0 minutes.

**36** Which of the following trends concerning Period 3 elements from Na to Cl are true?

**1** There is a change from metallic behaviour to non-metallic behaviour.

**2** Their compounds show an increase in the maximum oxidation number across the period.

**3** The melting points of the elements decrease across the period.

**37** Which of the following pairs of 0.1 mol $dm^{-3}$ aqueous solutions will not give a precipitate when added together?

**1** $FeSO_4$ and $MgCl_2$

**2** NaOH and $Ba(NO_3)_2$

**3** $Ba(OH)_2$ and $Cu(NO_3)_2$

**38** From which sodium salts would a precipitate be obtained on adding $AgNO_3$(aq) acidified with dilute $HNO_3$?

**1** fluoride

**2** chloride

**3** bromide

**39** Which of the following reactions will give identical observation?

**1** Bromine and ethene

**2** Bromine and ethane in the presence of uv light

**3** Bromine and benzene in the presence of halogen carrier

**40** Compound **X** has molecular formula $C_3H_4O_4$.

- It can react with HCl to give $C_3H_3O_3Cl$.
- One mole of X reacts with sodium metal to give one mole of $H_2$ gas.

Which of the following can be **X**?

**1** $CH_2(OH)COCOOH$

**2** CHOCH(OH)COOH

**3** $CH_2(COOH)_2$

**END OF PAPER**